

The Circle Theorems Reference Card

Every angle problem in this chapter is answered by ONE question: where is the vertex? Every length problem by one more: chords, secants, or a tangent?

Where is the vertex?	The formula	How to remember it
Vertex at the CENTER	angle = the arc	a central angle simply equals its intercepted arc
Vertex ON the circle (chords)	angle = $\frac{1}{2} \cdot \text{arc}$	the Inscribed Angle Theorem — half the intercepted arc
Vertex ON the circle (tangent + chord)	angle = $\frac{1}{2} \cdot \text{arc}$	same rule — half the arc the chord cuts off on that side
Vertex INSIDE the circle	angle = $\frac{1}{2} (\text{arc} + \text{arc})$	half the SUM of the two vertical arcs
Vertex OUTSIDE the circle	angle = $\frac{1}{2} (\text{far} - \text{near})$	half the DIFFERENCE — works for two secants, two tangents, or one of each
Circumscribed angle (two tangents)	angle = $180^\circ - \text{central}$	a shortcut form of the outside rule

The setup	The equation	Watch out
Two chords cross inside	(piece)(piece) = (piece)(piece)	multiply the two pieces of EACH chord
Two secants from outside	(whole)(external) = (whole)(external)	each product uses the WHOLE secant times just its outside part
Tangent + secant from outside	tangent² = (whole)(external)	a tangent is its own 'whole' and 'external' at once

The Two Mistakes That Cause Most Lost Points

- Using the near arc alone for an outside angle. Outside angles need the DIFFERENCE of two arcs, far minus near, then half.
- Multiplying (external)(far part) for secants. The product is (WHOLE)(external) — the whole secant is external + far part.

Chords, Tangents, and the Coordinate Plane

The rest of the chapter in four boxes — with the right-triangle habit that solves most of it.

Tangent Facts (10.1)

- Tangent $\perp$ radius at the point of tangency — this creates the right triangles behind most tangent problems.
- Two tangent segments from the same outside point are congruent: set the expressions equal.
- Find-the-radius trick: $\text{tangent}^2 + r^2 = (r + \text{external})^2$. The r^2 terms cancel and a linear equation is left.

Chord Facts (10.3)

- Congruent chords $\leftrightarrow$ congruent arcs $\leftrightarrow$ equidistant from the center (in the same or congruent circles).
- A diameter perpendicular to a chord bisects the chord AND its arc.
- The master right triangle: radius to the chord's endpoint = hypotenuse; half the chord and the distance to the center = legs. $r^2 = (\text{half-chord})^2 + d^2$.

Inscribed Polygons (10.4)

- Triangle in a semicircle $\rightarrow$ the angle opposite the diameter is 90° .
- Quadrilateral in a circle $\rightarrow$ opposite angles are supplementary (add to 180°).
- Same arc $\rightarrow$ same inscribed angle, no matter where the vertex sits on the far side.

Equations of Circles (10.7)

- $(x - h)^2 + (y - k)^2 = r^2$ — subtract the center's coordinates, so $(x + 3)^2$ means $h = -3$.
- Radius from a point: distance formula from the center. Diameter endpoints: midpoint gives the center first.
- Point tests: plug in — equal to r^2 is ON, less is INSIDE, more is OUTSIDE.

Name: _____

Which Rule? Practice

Name the rule — central angle, inscribed angle, tangent-chord, inside (sum), outside (difference), chords product, secants product, tangent-secant product, or circle equation. Then solve.

1. An inscribed angle intercepts a 156° arc. Find the angle.
2. Two chords cross; the vertical arcs are 70° and 110° . Find the angle.
3. Two secants leave a point; the far arc is 200° and the near arc 80° . Find the angle.
4. Chord pieces 5 and 6 meet pieces 3 and x . Find x .
5. A tangent t and a secant with external 4, far part 5. Find t .
6. Write the equation: center $(1, -3)$, radius 4.
7. A central angle measures 77° . Find its arc.
8. A tangent-chord angle is 63° . Find the intercepted arc.
9. A quadrilateral is inscribed; one angle is 88° . Find its opposite angle.
10. A triangle is inscribed with one side a diameter; one acute angle is 28° . Find the other.
11. Two tangents from a point cut arcs of 208° and 152° . Find the angle between them.
12. A chord of 48 sits 7 from the center. Find the radius.
13. Two secants: external 6 far 10; external 8 far x . Find x .
14. Is $(6, -8)$ on $x^2 + y^2 = 100$?
15. An inscribed angle is 44° . Find the central angle on the same arc.